4 WAYS TO GET HANDS-ON PRACTICE FOR YOUR CISCO EXAM

by

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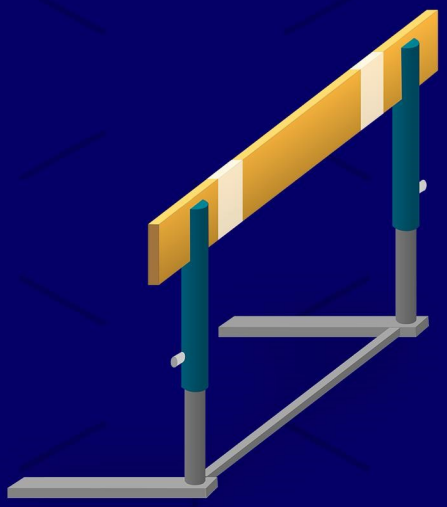
CCIEx2 #7945 EMERITUS

(ENTERPRISE INFRASTRUCTURE &COLLABORATION)

One of the biggest challenges in preparing for a Cisco exam, and one of the biggest hurdles to being truly qualified for a networking career, is getting hands-on experience with Cisco gear, without paying the thousands and thousands of dollars required to purchase new equipment from Cisco.



The good news is, in this article, I'll show you four free or inexpensive ways to get that critical hands-on experience, something I call "getting your hands dirty."

KEVIN WALLACE TRAINING,LLC

OPTION #1:
CISCO PACKET TRACER

If you're getting started with Cisco technologies, or perhaps preparing for Cisco's CCNA certification exam, look no further than a completely free tool Cisco generously makes available. It's called ***Cisco Packet Tracer***, and it's a network emulator.

Cisco Packet Tracer allows you to drag-and-drop multiple network components to construct a topology and then configure each of those components. And we're not just taking about routers and switches. The current version of Cisco Packet Tracer allows you to simulate more complex topologies, including wireless devices (e.g., wireless clients, wireless access points, and wireless LAN controllers), as seen in Figure 1.

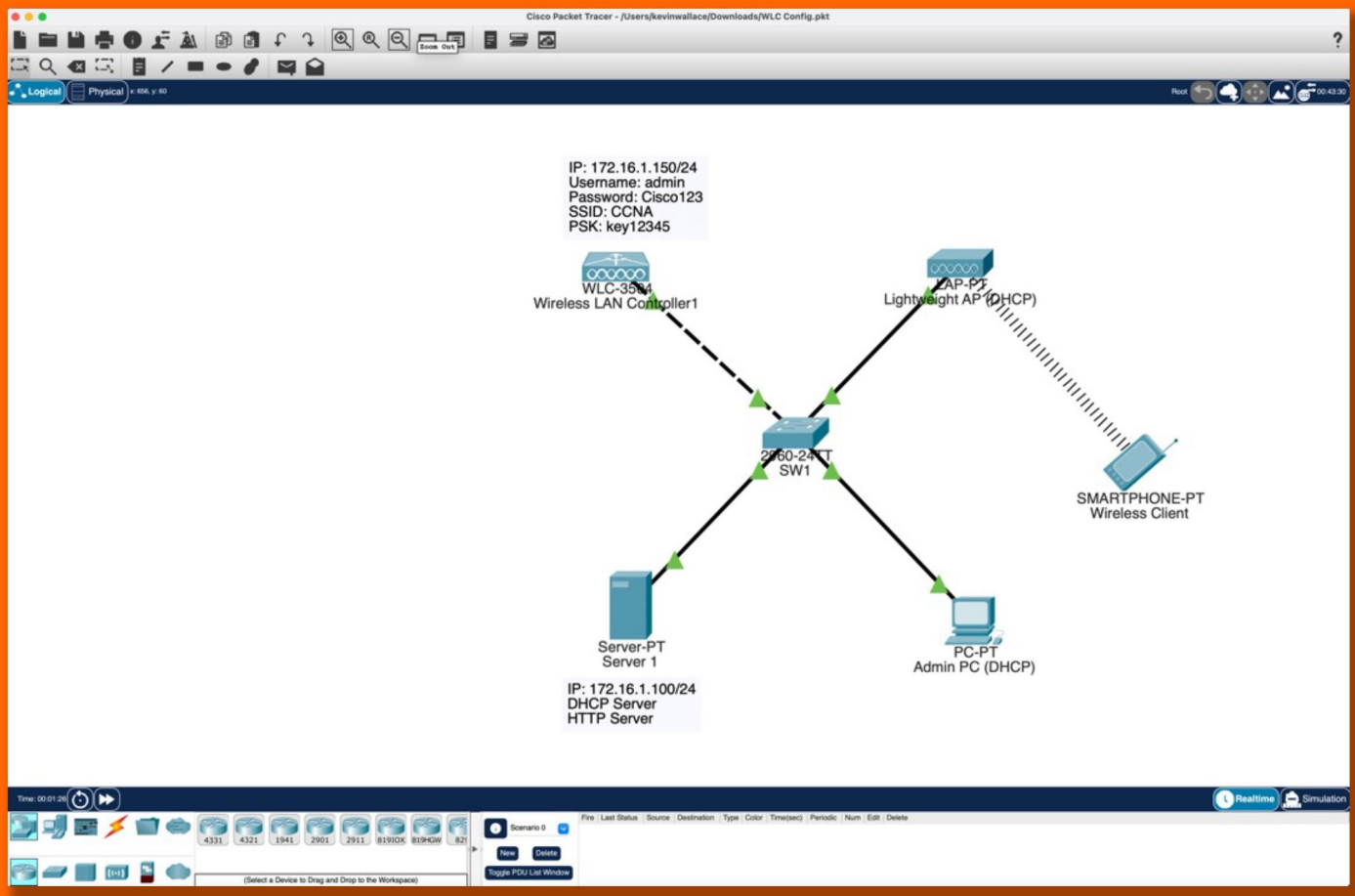


Figure 1 - A Wireless LAN in Cisco Packet Tracer

After you see a screen like the one shown in Figure 2, click the ***Get Started*** button. You'll then be asked to select your preferred language and click a ***Continue*** button.

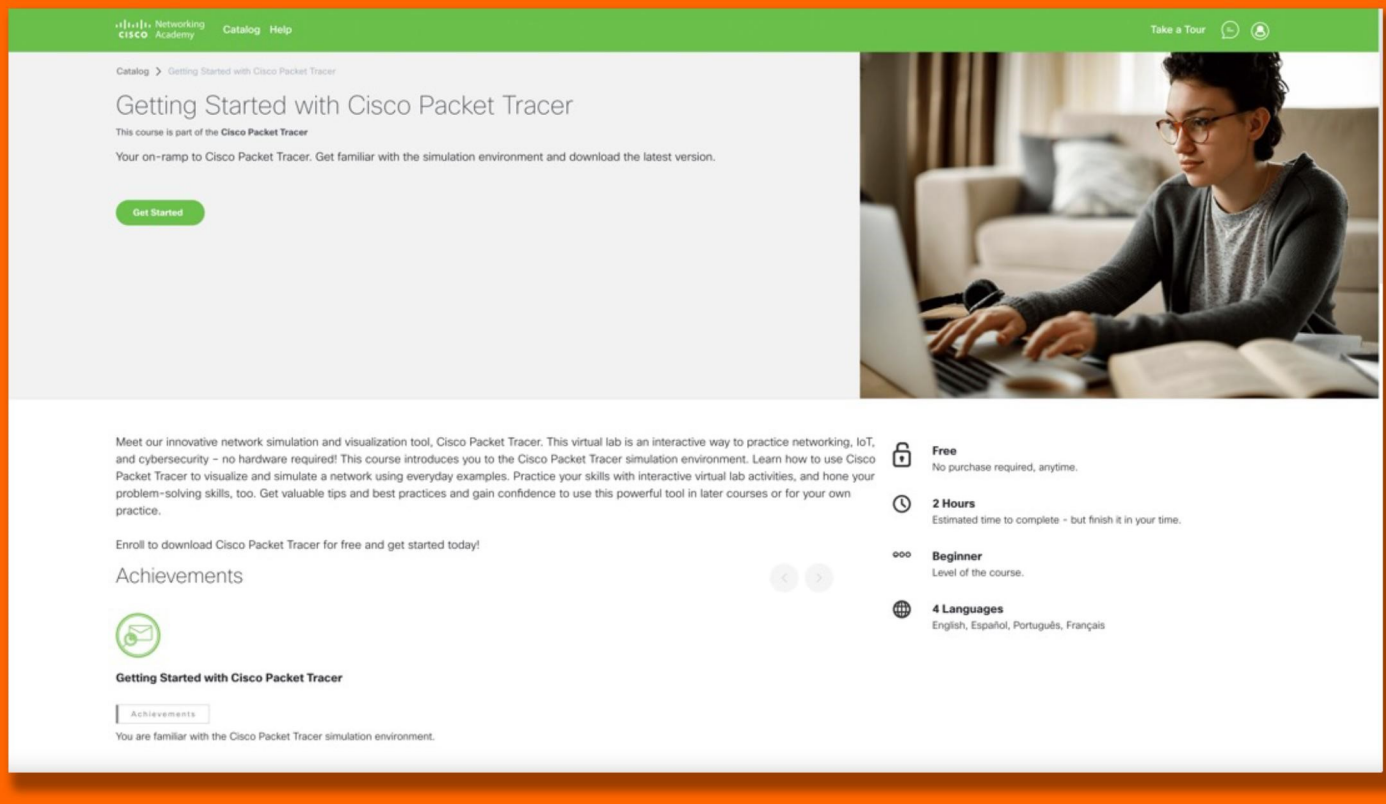


Figure 2 - Getting Started with Cisco Packet Tracer Initial Screen

Next, you'll need to login to your course, as shown in Figure 3. If you have an existing ***Google*** or ***Cisco Networking Academy*** account you can use one of those. Alternately, you can click the ***Sign up*** link to create a new account. You might then be asked to enter your country of residence, year of birth, and month of birth, as seen in Figure 3.

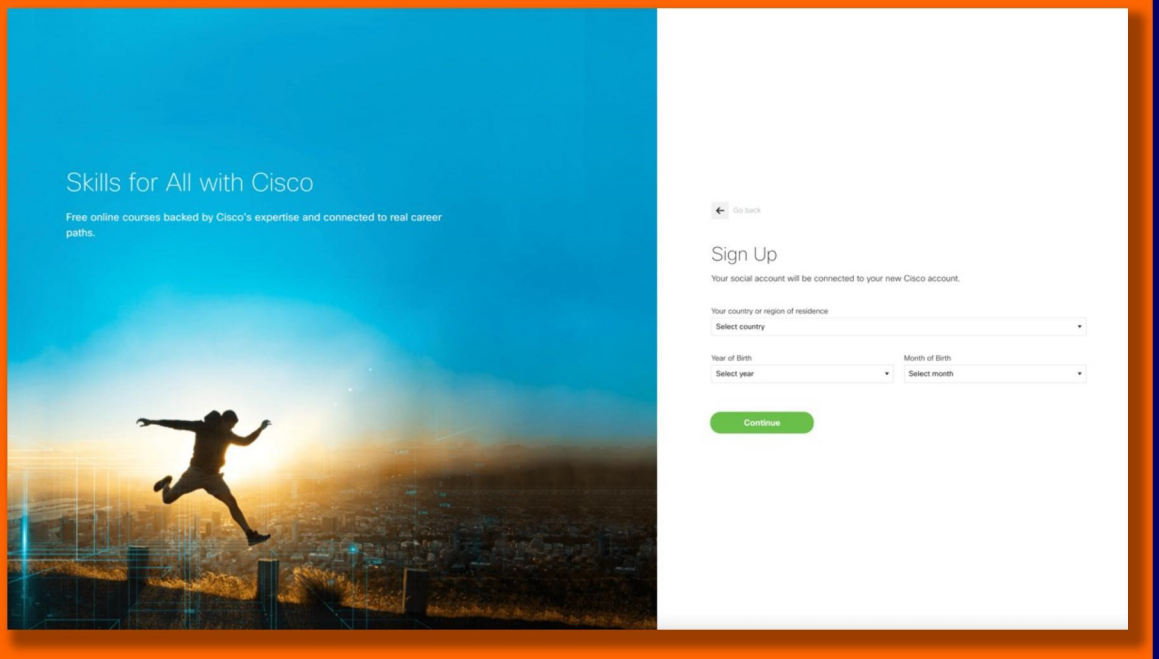


Figure 3 - Providing Sign Up Information

Then, you'll be asked to agree to Cisco Networking Academy's Terms and Conditions, after which you'll find yourself in Module 1 of your free course, as seen in Figure 4.

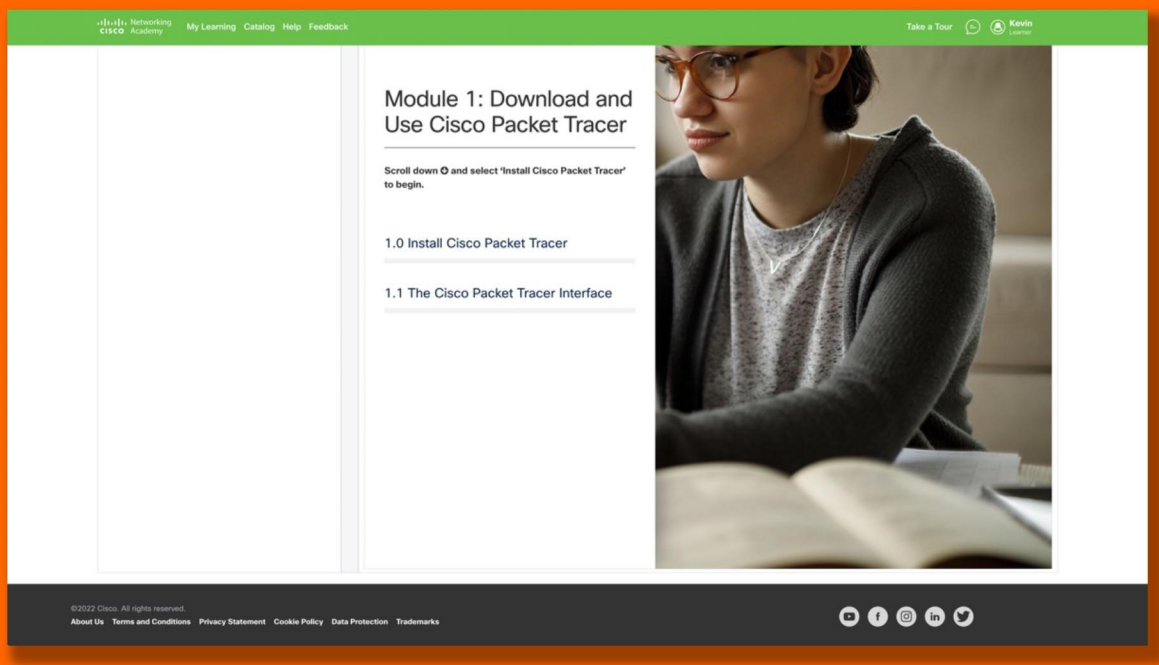


Figure 4 - Module 1 of Your Course

From there, you'll want to click on ***1.0 Install Cisco Packet Tracer*** and scroll down the next page until you reach the ***1.0.3 Download Cisco Packet Tracer*** section, as seen in Figure 5.

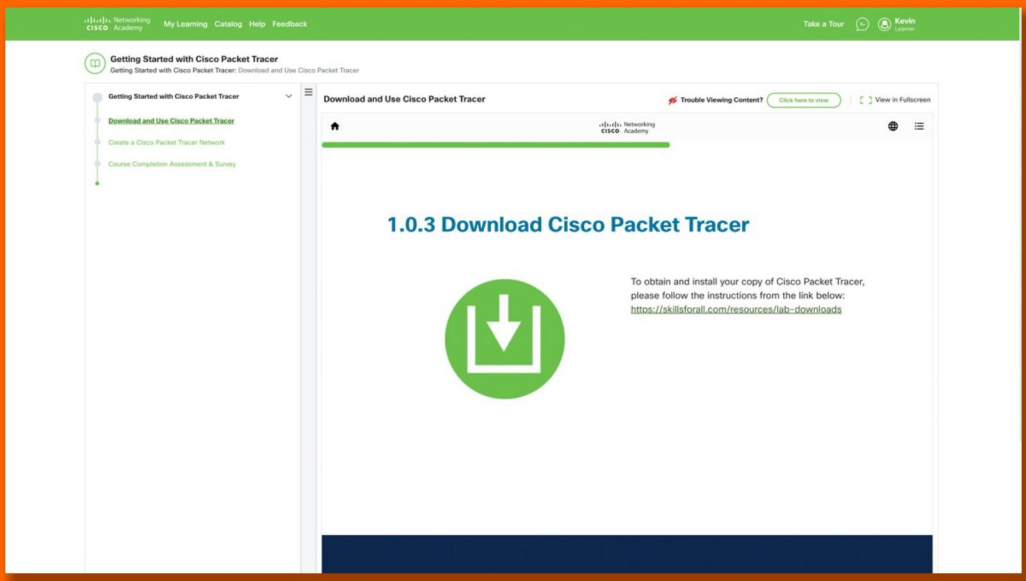


Figure 5 - The Download Section

You can then click the link provided to navigate to the Resource Hub. Finally! You're on the download page, as seen in Figure 6. Select the version of Cisco Packet Tracer you want to install. At the time of this writing, versions are available for MacOS, Ubuntu, and Microsoft Windows.

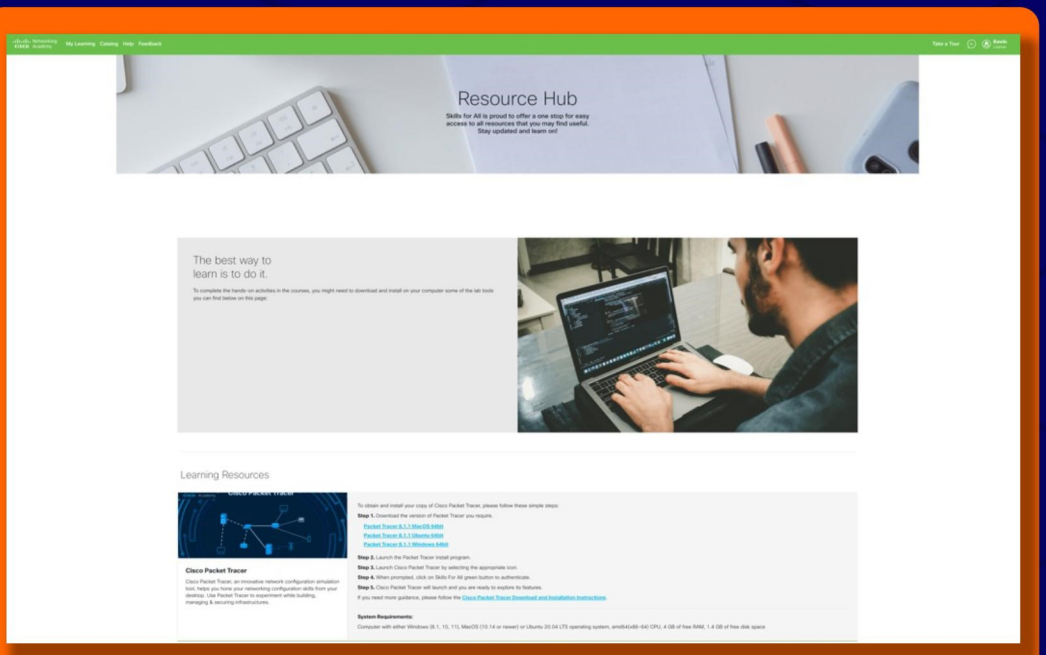


Figure 6 - The Resource Hub

By the way, as long as you're enrolled in the Cisco Networking Academy's ***Getting Started with Cisco Packet Tracer*** course, I recommend going through the course to get as much as you can from this amazing free application from Cisco.

OPTION #2:
CISCO MODELING LABS (CML)

While I typically recommend Cisco Packet Tracer to my CCNA students, I recommend ***Cisco Modeling Labs*** (CML) to my CCNP students. Unlike Cisco Packet Tracer, which is a simulating Cisco IOS, CML, as seen in Figure 7, is an *emulator*, running actual Cisco IOS on virtualized devices. Just be aware that while Cisco Packet Tracer is free, CML is not (if you want to run it on your computer… more on that later).

Like Cisco Packet Tracer, you can drag-and-drop a variety of network devices to construct your own topology. However, being an emulator, CML lets you perform advanced lab scenarios not possible on Cisco Packet Tracer.

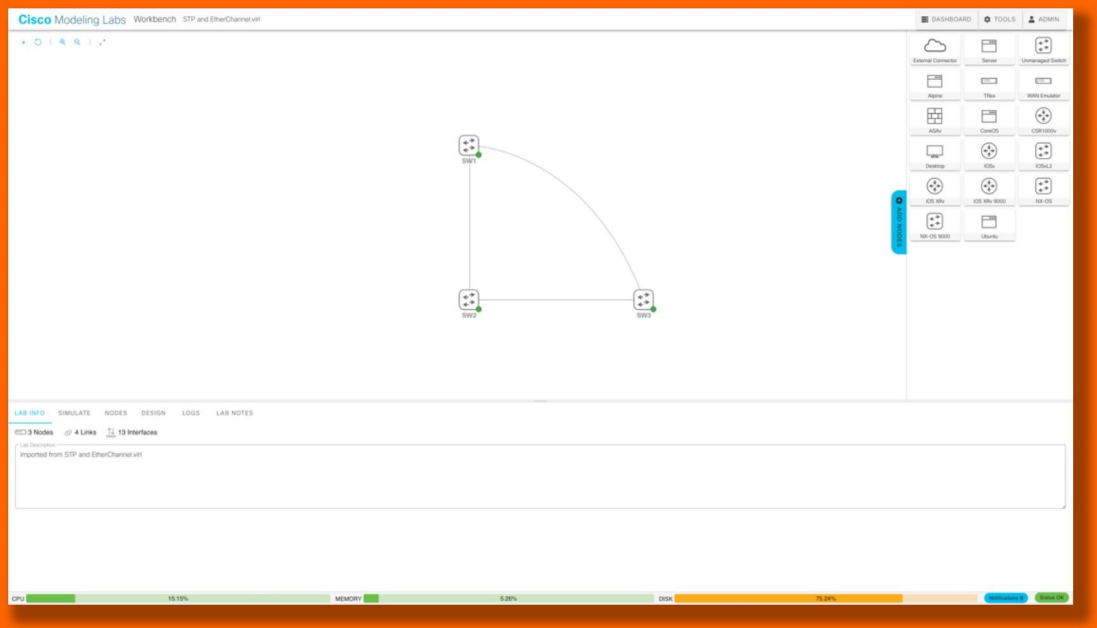


Figure 7 - Cisco Modeling Labs (CML)

Another contrast is CML doesn't install as an application on your computer. Rather, it runs in a virtual machine. At the time of this writing, Cisco lists the following as supported virtual machine hypervisors:

- VMware Workstation 14 or later
- VMware Fusion Pro 10 or later
- VMware Player 14 or later
- VMware ESXi 6.5 or later

Personally, I've run CML on *VMware Fusion Pro* on my Mac and on *VMware ESXi* on an external server I have that runs several other virtual machines. You can purchase CML from The Cisco Learning Network Store using the following link:

https://learningnetworkstore.cisco.com/cisco-modeling-labs-personal

At the time of this writing, there are two options available for purchasing CML. You can choose the *Personal* version (at a price of $199) or the *Personal Plus* version (at a price of $349), as seen in Figure 8.

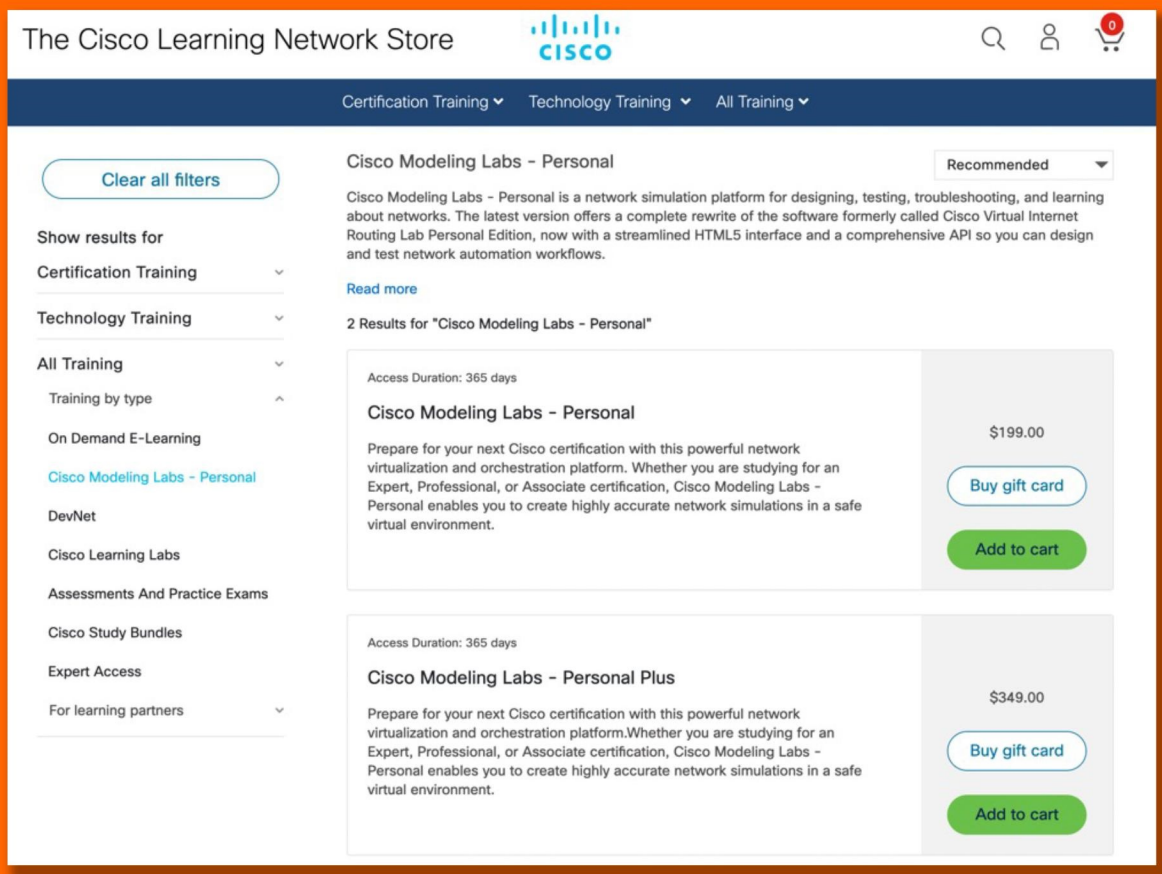


Figure 8 - CML Personal vs. Personal Plus Versions

Either purchase gives you 365 days of access to the software. The big difference is the number of nodes (e.g., routers and switches) supported. The Personal version supports 20 nodes, while the Personal Plus version supports 40 nodes.

For CCNP studies, I recommend the Personal version. For CCIE studies, however, I recommend the Personal Plus version.

OPTION #3:
CISCO'S DEVNET SANDBOX

The folks at Cisco DevNet did a huge favor for us with the creation of their Cisco DevNet Sandbox. Specifically, the *Cisco DevNet Sandbox* offers us a plethora of lab topologies, which we can connect to remotely. These topologies span a wide range of Cisco technologies, including basic networking, collaboration, Internet of Things (IoT), data center, cloud, security, and open source, as seen in Figure 9.

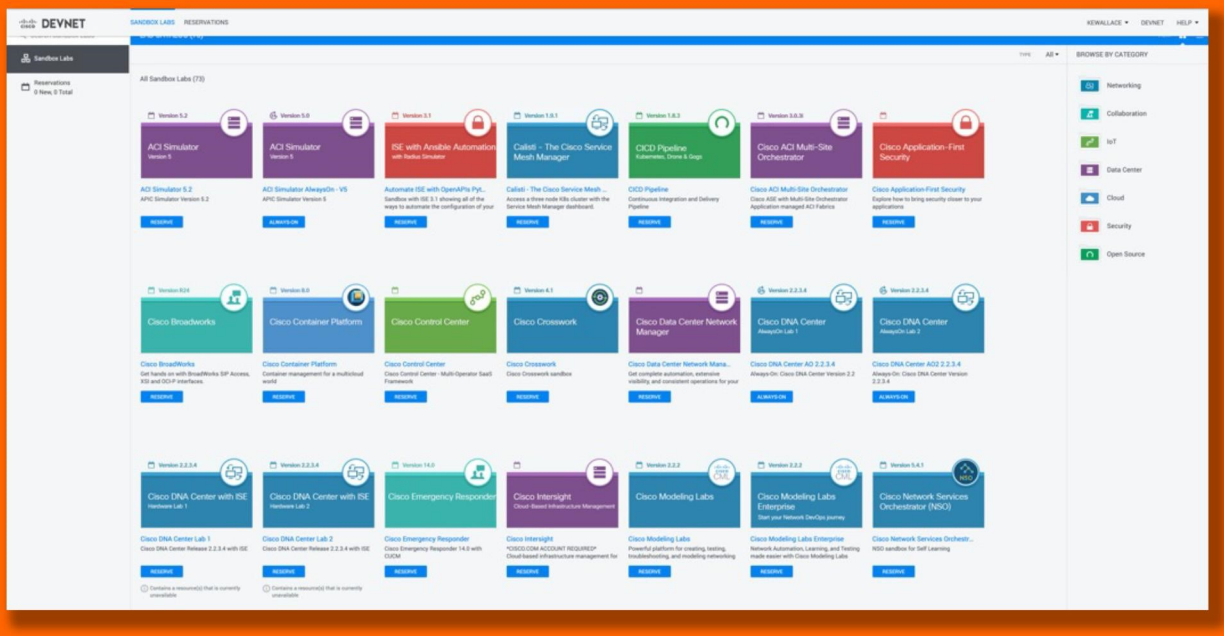


Figure 9 - Cisco's DevNet Sandbox

To access this sandbox environment, you can visit https://developer.cisco.com/site/sandbox and login with a variety of credential options. Some lab topologies can be reserved for a specific data and time and for a specific duration, while other topologies are in an "Always-On" state that allows you to connect without a reservation. (**NOTE:** The Always-On lab topologies are typically read-only, and you cannot make any configuration changes.)

Something else to be aware of is the DevNet team is continually updating the lab topologies in their sandbox. So, it's likely the lab options you seen when visiting their site are different than those seen in the previous figure.

However, at the time of this writing, one of the lab topologies we can reserve is (drumroll please)… *Cisco Modeling Labs*! That's right… if you'd rather not pay for CML, you can get access to it here for FREE! The downside is, you have to make a reservation and setup a VPN connection to the sandbox every time you want to use it. But, if you save nearly $200 a year, that might be an acceptable tradeoff.

Let's take a look at how we can access CML for free using the Cisco DevNet Sandbox. After we connect to the sandbox (using the previous URL) and login (using one of multiple credential options), we can click the RESERVE button under Cisco Modeling Labs, as seen in Figure 10.

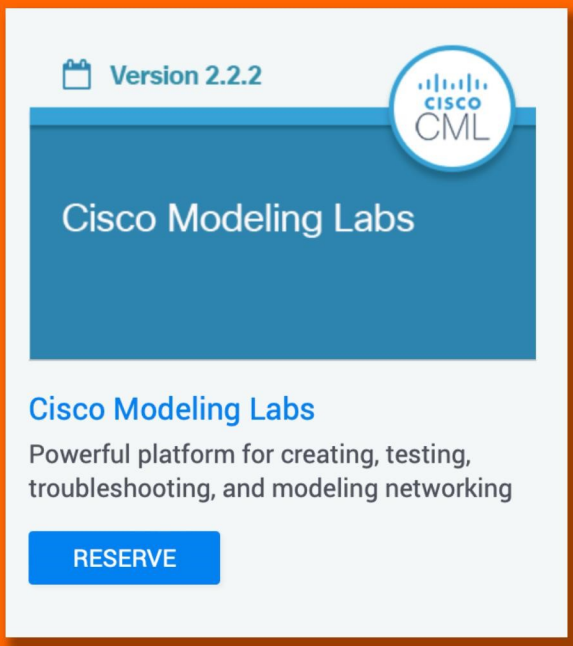


Figure 10 - Reserving CML

You can reserve CML for a maximum of four hours, but the default reservation time is two hours, as seen in Figure 11.

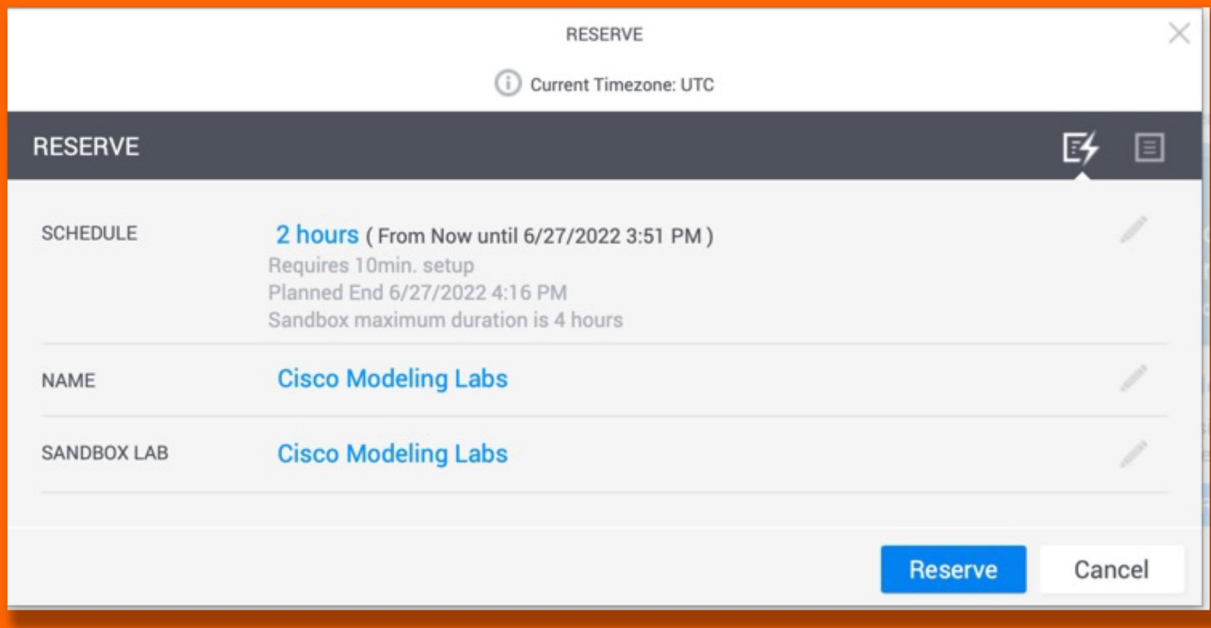


Figure 11 - Default Reservation Settings

If you want to increase the reservation time to four hours, click the pencil icon in the Schedule section to edit the duration, as seen in Figure 12.

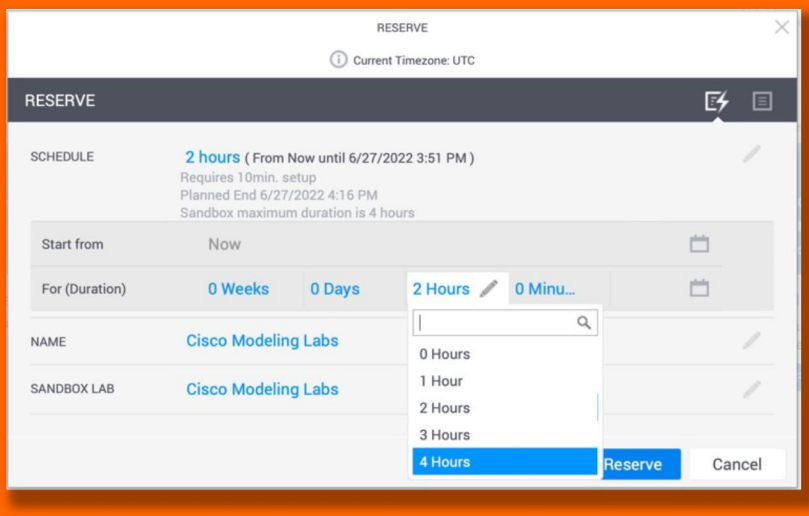


Figure 12 - Setting Reservation Duration

To access the CML sandbox (and most of the other sandboxes), you'll need to setup a VPN connection. Fortunately, after you request your reservation, you'll be taken to a topology of your lab, along with three tabs providing instructions and information. If you click on the VPN Access tab, as seen in Figure 13, you'll find links to download the ***Cisco AnyConnect VPN Client*** software, along with instructions on installation and using Cisco AnyConnect to connect to your lab.

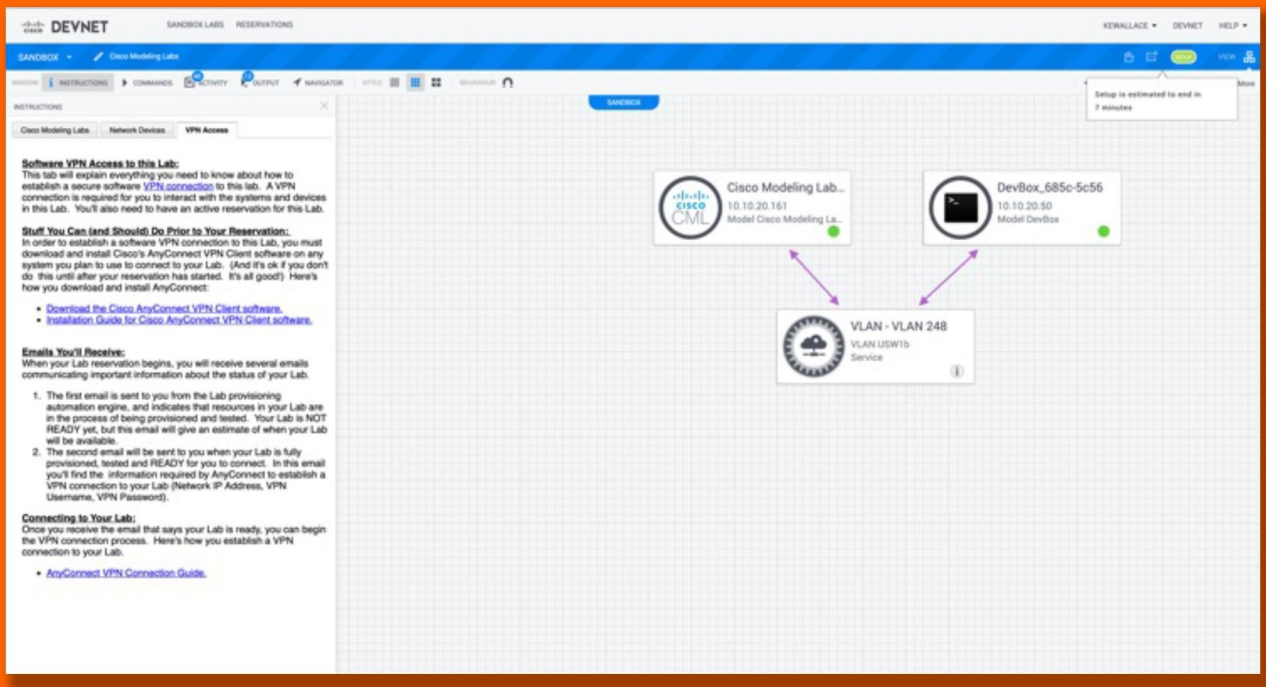


Figure 13 - VPN Access Tab

Once your lab topology has initialized and your VPN connection is established, it's time to log into the CML interface (using the IP address and login credentials shown on the Cisco Modeling Labs tab), as seen in Figure 14.

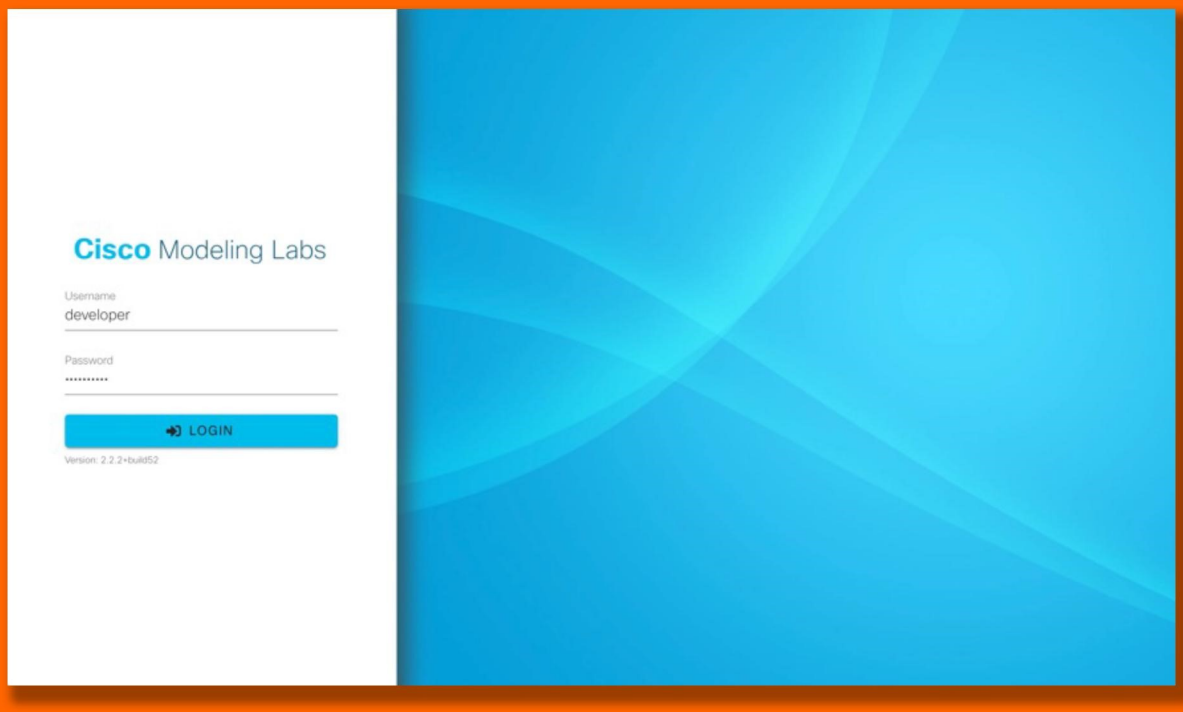


Figure 14 - CML Login

Once you login, you'll notice there's already a topology (named *Multi Platform Network*) up and running, as seen in Figure 15.

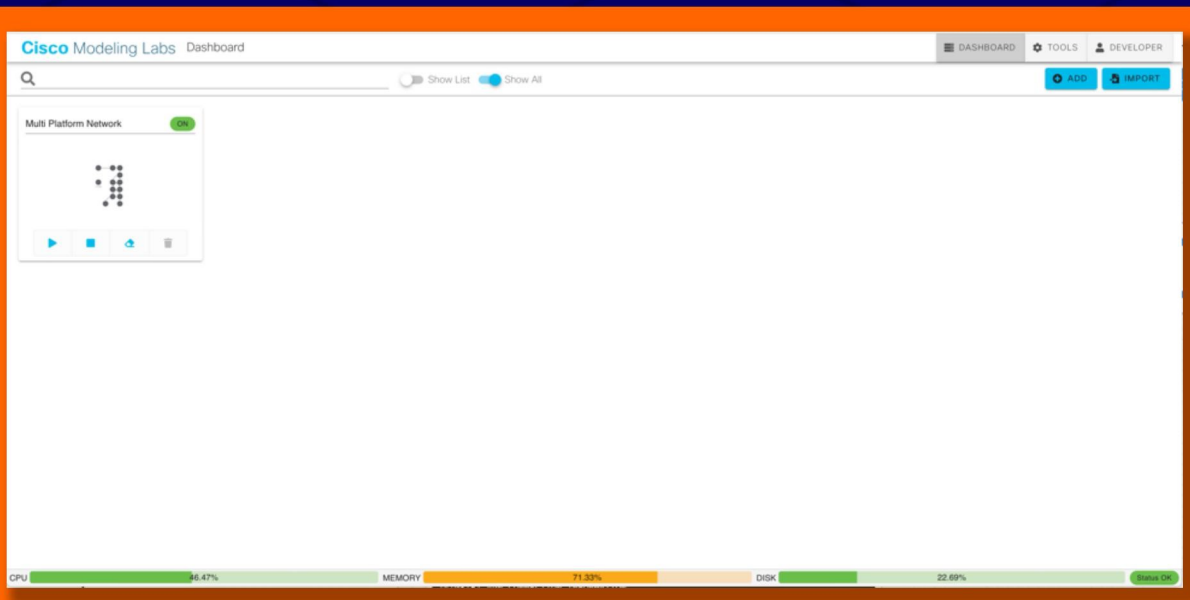


Figure 15 - Default Sandbox CML Topology

While you could certainly explore this very robust default topology, you might instead prefer to create your own topology. So, what you can do is stop the Multi Platform topology by clicking the ***Stop*** button (i.e., the blue square) in the topology pane. Then, you can click the ***ADD*** button to create your own topology, as seen in Figure 16.

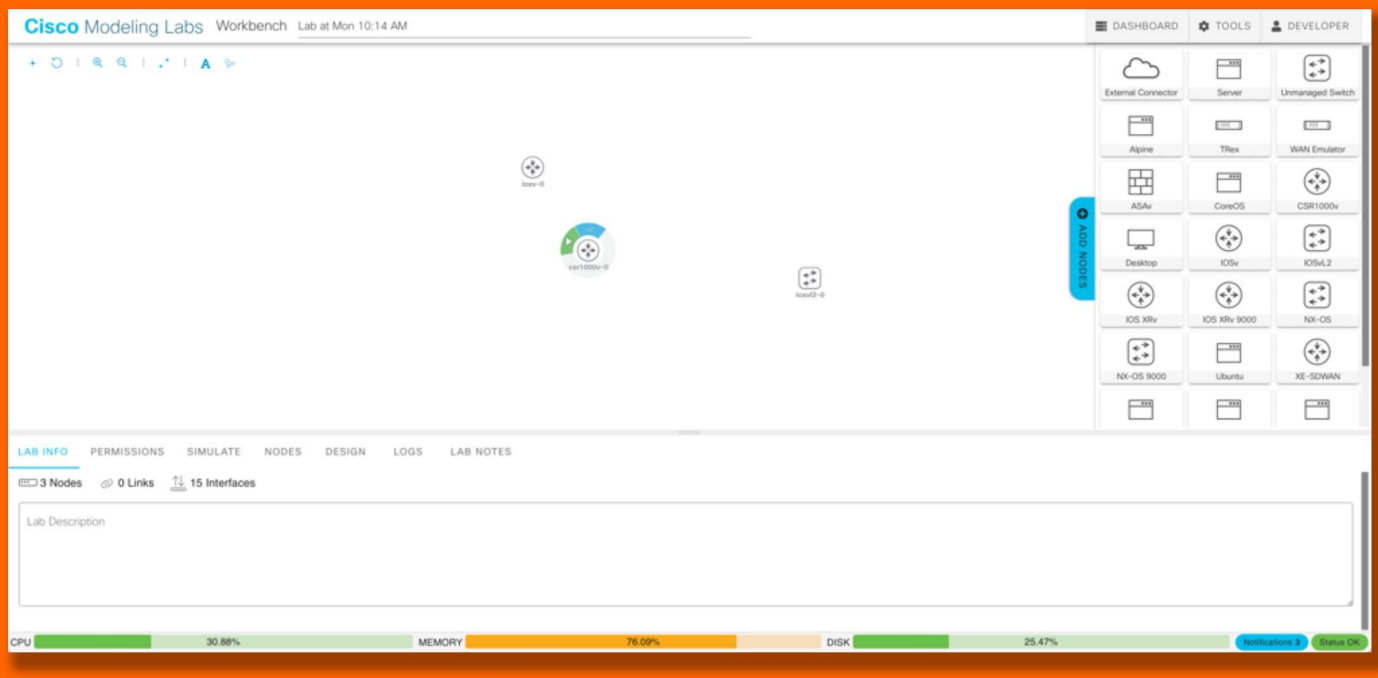


Figure 16 - Creating Your Own CML Topology

Admittedly, that's a lot of steps to go through. So, below is a link you can use to check out a video tutorial on my YouTube channel what walks you through accessing CML for free using Cisco's DevNet Sandbox.

https://bit.ly/sandboxcml

I encourage you to go through the process of setting up CML in the DevNet Sandbox environment a couple of times. Once you get the hang of it, this option might be your go-to solution for labbing up various topologies.

OPTION #4:
BUYING USED GEAR

Another option to get your hands dirty is to purchase used gear, which you can find on a variety of websites. Working with actual gear gives you the advantage of getting familiar with the cabling and hardware you'll be working with in the real world. On the downside, this might be the most expensive of the four options given here.

Personally, I usually head over to eBay when looking for used Cisco gear, as seen in Figure 17.

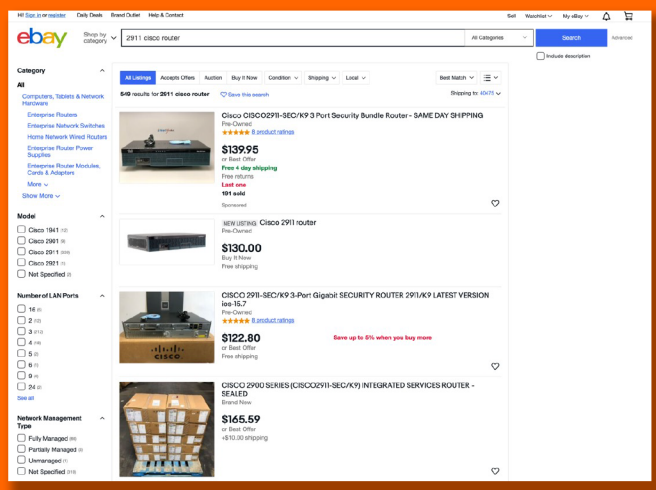


Figure 17 - Purchasing Used Gear on eBay

But the big question I get from my students is, "what should I buy?" While there many combinations of routers and switches that would get you a long way in your CCNA or CCNP Enterprise studies, my general recommendation is the following:

- Cisco 2911 Router – Quantity: 3
- Cisco 3560 Catalyst Switch – Quantity: 1
- Cisco 2960 Catalyst Switch – Quantity: 2

This combination of routers and switches will allow you to construct several labbing scenarios, such as multi-area OSPF, redistribution, STP, PBR, SVIs, and lots more.

Also, as you're considering the pros and cons of purchasing real gear, realize that you might be able to sell the gear once you're finished with it. Personally, when I went after each of my CCIE certifications, I purchased a collection of routers, switches, and other gear from eBay. Then, after passing each of my labs, I sold the equipment. Each time, I was able to sell the equipment for nearly the amount I originally paid. So, please keep in mind that the option of reselling your gear might help mitigate the initial investment.

COMPARISON

Now that we've considered four ways of getting hands-on experience for your Cisco exam, let's do a quick recap by referring to the following table:

OPTION	PROS	CONS
CISCO PACKET TRACER	• Free • Allows you to create a lab topology with many devices • Easy to install	• Simulates, rather than emulates, Cisco IOS, resulting in some features not being supported
CISCO MODELING LABS (CML)	• Runs actual Cisco IOS, rather than emulating it • Allows you to create a lab topology with as many as 20 devices (with the Personal version)	• Cost (currently $199 for one year of access to the Personal version) • Slightly more challenging to configure than Cisco Packet Tracer (requiring the use of a hypervisor, such as VMware ESXi)
CISCO DEVNET SANDBOX	• Free • Allows reservations to CML • Allows reservations for many other lab topologies spanning a wide range of technologies	• Requires VPN configuration • Configurations must be exported to be used in the future • Desired reservation dates or times might not be available
PURCHASING USED GEAR	• Provides a true hands-on experience	• Most expensive option (in up-front cost)

SUMMARY

Did you get some clarity about options for getting your hands dirty with Cisco gear? I sincerely hope so. Also, as a "*Thank you!*" for checking out our guide, you're invited to sample the first module of one of our video courses, absolutely free!

Just visit the following link to get your Module 1 from any of these courses for FREE!

kwtrain.com/course-samples

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